Tracking Political Themes in the French Fifth Republic: A Dynamic Topic Modeling Approach using BERTopic

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Abstract

This project investigates the temporal evolution of French political discourse by analyzing 9,627 legislative electoral manifestos from the Archelec corpus across four election years (1973, 1978, 1981, and 1988). To track these semantic shifts, we first conduct a comparative analysis between Latent Dirichlet Allocation (LDA) as baseline, and BERTopic, a modern neural approach using Transformer embeddings. While LDA successfully captures ideological divides, such as the left-right spectrum, it struggles with overlapping clusters and dense semantic spaces. In contrast, BERTopic seems to demonstrate superior qualitative performance by isolating specific thematic issues such as ecology, European debates, and Breton regionalism that LDA fails to cleanly detect. Furthermore, by applying a Dynamic Topic Modeling framework, BERTopic is able to map the emergence or the decline of these themes over time. However, traditional quantitative coherence metrics evaluate LDA slightly more favorably than BERTopic even if it seems to have better qualitative results. This highlights the "validation gap" in evaluating neural models using metrics with a structural bias toward toward exact word co-occurrences. Ultimately, this study confirms BERTopic's enhanced capability for extracting highly interpretable and dynamic political narratives.

Introduction

Topic modeling is a powerful unsupervised technique for discovering latent semantic structures within large document collections. Moreover extending this approach across time allows the analysis of how these semantic representations evolve over time.

This project investigates the evolution of political discourse using the Archelec corpus, a comprehensive collection of electoral manifestos from candidates in the French Fifth Republic, digitized by Sciences Po. By analyzing several years of legislative campaigns the goal is to track and visualize the temporal shifts in political themes.

To achieve this, I conduct a comparative analysis between a traditional statistical baseline: Latent Dirichlet Allocation (Blei et al. [2003]) and BERTopic (Grootendorst [2022]), a neural approach leveraging Transformer embeddings.

This will lead to the following questions: How have French political themes evolved across different election years? and to what extent do modern models extract more interpretable topics compared to classical methods especially given the challenges of automated coherence evaluation?

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

1 Data Description

I used a subset of the Archelec corpus comprising 9 627 manifestos from the legislative elections of 1973, 1978, 1981, and 1988 distributed as illustrated in Figure 1.
In order to improve the models, the manifestos have to be preprocessed. I used Spacy in order to eliminate french stopword and lemmatized the text, which reduces words to their base or dictionary form. This step is essential to unify semantic concepts, decrease the sparsity of the vocabulary and ensure that the models group underlying themes rather than grammatical variations.

Figure 3 shows the distribution of document lengths after preprocessing. Across the selected corpus, the mean text length is 4 386 characters with a median of 3 876.

Figure 2 illustrates the distribution of identified political parties within the dataset. I could have aggregated candidates into bigger political coalitions (such as "left-wing," "right-wing," or "center") to facilitate latent space projections if needed but a vast majority of candidates lack a identified group affiliation. Reconstructing these affiliations manually falls outside the scope of this study as the main objective focuses on global shifts in French political discourse over time rather than intraparty dynamics.

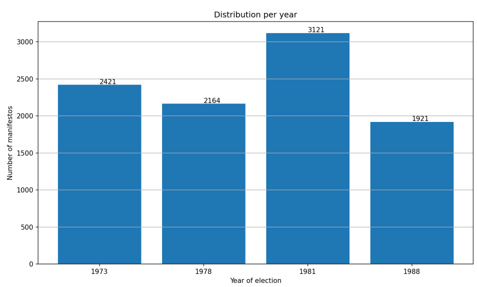

Figure 1: Distribution of documents across the years

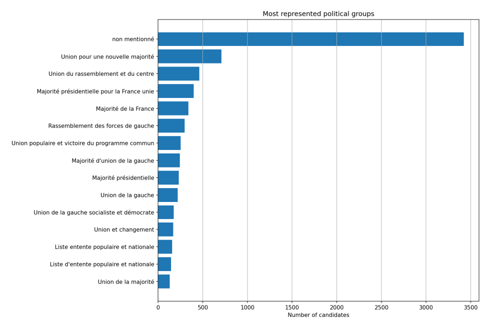

Figure 2: Distribution of poltical groups in the corps

2 Models and evaluation

Reminder of LDA Latent Dirichlet Allocation (LDA) was first introduced by Blei et al. [2003] and is based on 3 hypothesis :

- The number of topic is known and fixed
- The order of the words does not matter (bag of word)
- The order of the document does not matt

That means that documents and words are observed but topics are hidden. One default of this method is that we need to select the number of topics before running the model.

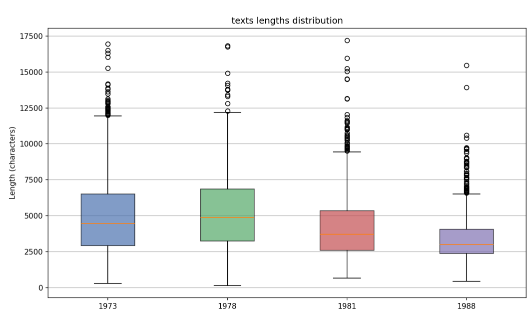

Figure 3: Distribution of text lengths

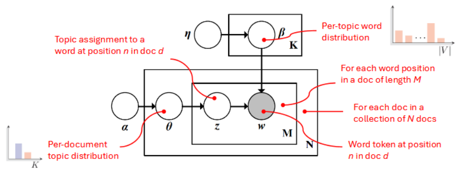

Figure 4: LDA scheme

The diagram 4 illustrates the probabilistic generative process of LDA. The rectangles represent repeated elements: K is the total number of possible topics, N represents the number of documents and M represents the word position within a specific document. White circles represent hidden variables while the w circle represents the observed data (word token at position n in document d).

It begins with the Dirichlet priors α and η. The parameter α governs the per-document topic distribution, θ defining the proportion of topics within document d. Concurrently, η governs the per-topic word distribution, β, which defines the probability of words occurring within a specific topic. To generate a specific word w, the model first draws a topic assignment z from the document's topic distribution θ. Then, the final word w is drawn from the corresponding topic's word distribution β. In practice, the goal of the LDA algorithm is to reverse this process: using the observed words w to automatically infer the hidden thematic structure—the topics and their proportions—across the entire collection of documents.

BERTopic Transformers have brought new models which are more adapted for topic modeling and BERTopic Grootendorst [2022] is one of them. BERTopic approaches topic modeling as a clustering task in a semantic space. After generating and reducing document embeddings, all documents within the same cluster c are concatenated to form a single document representing that class. To extract the representative word distribution for this topic, BERTopic introduces c-TF-IDF: a class-based variation of the classical TF-IDF, calculated as follows: $W_{t,c} = tf_{t,c} * log(1 + \frac{A}{tf_t})$

Where $tf_{t,c}$ is the frequency of term t in cluster c. Since it is needed that the word appears a lot in cluster c to be a represent of the cluster. The more the word is used, the bigger is $tf_{t,c}$.
The inverse class frequency is calculated by taking the logarithm of A, the average number of words per class, divided by tf_t, the frequency of term t across all classes. That penalizes words that are frequent in each cluster, because if a word is common, the $tf_{t,c}$. The addition of 1 within the logarithm ensures that the function outputs only positive values. This models the importance of words in clusters instead of individual documents and generates the final topic-word distributions for each cluster.

This model has the advantage that it does not require to specify the number of topics (cluster) and is intelligent to gather document since it does not use bag of words but transformer to include the context. Moreover it is possible to follow the evolution of topics across the time based on a c-TF-IDF method. The main hypothesis is that a global topic exists independently of time even if its specific vocabulary evolves.
First it is needed to generate a global representation of topics by fitting the model on the entire corpus ignoring any temporal aspects. Then the following formula allow us to have the most representative words for a given year. $W_{t,c,i} = tf_{t,c,i} * log(1 + \frac{A}{tf_t})$

Evaluation Evaluating the quality of topic models is subjective when it is done manually by humans. To provide quantitative assessment, it is possible to use topic coherence metrics. These metrics measure whether the top words associated with a given topic are semantically related and make sense when grouped together. I used three different coherence metrics to compare LDA and BERTopic: U_{Mass}, $NPMI$, and C_v.

U_{Mass}**:** was introduced in Mimno et al. [2011] is a metric which measures word co-occurrence based on document frequencies within the corpus. For the top N words of a topic it is formulated as:

$$C_{UMass} = \frac{2}{N \cdot (N-1)} \sum_{i=2}^{N} \sum_{j=1}^{i-1} \log \frac{P(w_i, w_j) + \epsilon}{P(w_j)}$$

(as it is written in Röder et al. [2015]).
Because it is the logarithm of probabilities, U_{Mass} scores are strictly negative. A score closer to 0 indicates a highly coherent topic, while lower negative values indicate worst coherence.

$NPMI$ (Normalized Pointwise Mutual Information) was introduced in Lau et al. [2014]. It measures the strength of association between pairs of words. It is calculated over a sliding window across the corpus.

$$NPMI(w_i, w_j) = \frac{\log \frac{P(w_i,w_j)}{P(w_i)P(w_j)}}{-\log P(w_i, w_j)}$$

A score of 1 indicates that the words perfectly co-occur, 0 implies independence, and -1 indicates they never appear together..

C_v was introduced in Röder et al. [2015] It is a metric that creates context vectors for each word using NPMI scores and then computes the average cosine similarity between these vectors:

$$c_v = \frac{\sum_{k=1}^{K} \sum_{n=1}^{N} s_{cos}(\vec{w}_{n,k}, \vec{w}_k^*)}{N \times K}$$

C_v scores range from 0 to 1. A higher score indicates excellent topic quality.

However, recent research highlights the need for caution when relying on automated coherence metrics to compare classical and neural models. Hoyle et al. [2021] demonstrated a significant "validation gap": human judgment differs from automated metrics. Therefore the automated evaluation of this topic modeling should ideally be complemented by human evaluation.

3 Methodology and Implementation

LDA Baseline A classical Latent Dirichlet Allocation model serves as the baseline trained on the entire merged corpus. Following the lemmatization process, document representations are generated using a bag-of-words approach. We utilize `CountVectorizer` with the vocabulary size restricted to $V = 1,000$ (`max_features`) to mitigate high dimensionality and computational overhead. To avoid noise, words appearing in fewer than 5 documents (`min_df` = 5) or in more than 90% of the documents (`max_df` = 0.9) are excluded. For this baseline, the number of topics is strictly set to $K = 10$ to establish a controlled environment for comparison.

BERTopic Rather than relying on discrete term frequencies, documents are embedded into a dense vector space using the `paraphrase-multilingual-MiniLM-L12-v2` Sentence-Transformer.

Once the high-dimensional embeddings are generated, UMAP is utilized for dimensionality reduction followed by HDBSCAN for clustering. To ensure a fair comparative analysis with the LDA baseline, the clustering algorithm is constrained to have exactly $K = 10$ topics (`nr_topics` = 10).

The BERTopic pipeline is executed in two phases. First, a global static model is trained on the entire corpus ignoring the temporal dimension, to learn the whole thematic structures of the Fifth Republic. Second, the Dynamic Topic Modeling framework is applied by computing the temporal c-TF-IDF for each election year. This allows us to track shifts and semantic variations in the vocabulary of each cluster over time.

Finally, the top representative words for each cluster are extracted for the evaluation. The outlier cluster (identified as Topic -1), which HDBSCAN inherently uses to isolate noisy and unclustered documents, is deliberately excluded from the evaluation to prevent bad coherence metrics.

4 Results

Figure 5 shows the top words for the 10 LDA topics generated from the entire corpus. An LLM was used to give an interpretation of these clusters. The synthesis is presented in Table 1:

First we can notice that there exist some bilingual manifestos since the last topic consists entirely of german stop words (likely due to the Alsace Lorraine regions). It could be handled by adding the german stop words to our Spacy preprocessing pipeline, but it does not prevent the analysis of topic modeling in our case since we just have to analyze the first 9 topics.

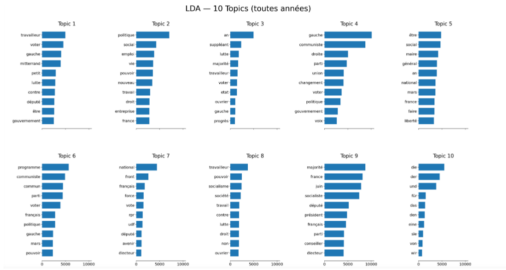

Figure 5: Top words for 10 clusters in the LDA an the whole corpus

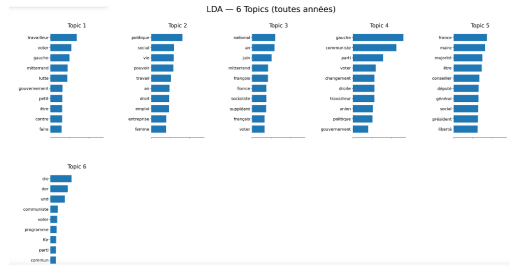

Figure 6: Top words for 6 clusters in the LDA an the whole corpus

Table 1: Synthesis of top words and topics for LDA via a LLM (in french)

Topic	Top words	Interpretation
Topic 1	travailleur, voter, gauche, mitterrand, lutte, député	Gauche mitterrandiste — campagne PS, monde du travail
Topic 2	politique, social, emploi, vie, pouvoir, travail, droit	Politique sociale généraliste — emploi, droits sociaux
Topic 3	an, suppléant, lutte, majorité, travailleur, état	Institutions & lutte politique — thèmes électoraux classiques
Topic 4	gauche, communiste, droite, parti, union, changement	Clivage idéologique gauche/droite — union de la gauche, PCF
Topic 5	être, social, maire, général, national, mars, france	Institutions locales/nationales — thème vague, mélange mairie/nation
Topic 6	programme, communiste, commun, parti, voter, politique	Programme commun / PCF — Programme commun de la gauche
Topic 7	national, front, français, force, vote, rpr, udf	Droite et extrême-droite — Front National, RPR, UDF
Topic 8	travailleur, pouvoir, socialisme, société, travail, lutte	Gauche radicale / lutte des classes — socialisme, ouvriers
Topic 9	majorité, france, juin, socialiste, député, président	Élections & institutions — résultats législatifs, présidence
Topic 10	die, der, und, für, das, den	Textes en allemand (documents alsaciens/bilingues dans le corpus)

Then we can see that many of these topics are about the political left (topic 1,6,8) which means that it is hard for LDA to discriminate between these nuances. This overlap becomes apparent with the pyLDAvis visualization (projection of the topic in a 2D space) figure 7 (visible in the git repo in `outputs\05_lda_interavtice.html`). This could result from having too many topics or an overly homogeneous vocabulary. To exclude the first hypothesis we can reduce the model to 6 topics. But one can observe more or less a similarly overlapping results (Figure 6 shows the LDA top word in this case). Some other topics are more about the left/right opposition such as topic 4 and 7. We can also notice that all topics have more or less the same weight which means that there is no major topic.

Figure 8 shows the top words for the 10 topics generated by BERTopic (except we removed the outlier topic). An LLM was once again used to interpret these clusters, as summarized in Table 2.

Topic 1 in BERTopic is relatively similar to Topic 1 in LDA. Topic 2 captures a more radical left-wing discourse. Topic 3 (Ecology), Topic 7 (Europe), and Topic 8 (Brittany regionalism) are highly specific thematic clusters that were entirely absent from the LDA results. Additionally, the "German" artifact topic from LDA is no longer present (likely having been absorbed into the outlier cluster (-1) by HDBSCAN). To conclude this initial comparison, BERTopic successfully "discovers" three

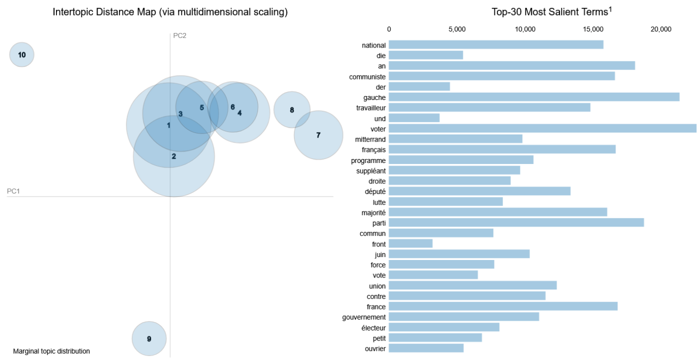

Figure 7: Projection of LDA topics in 2D via PyLDAvis

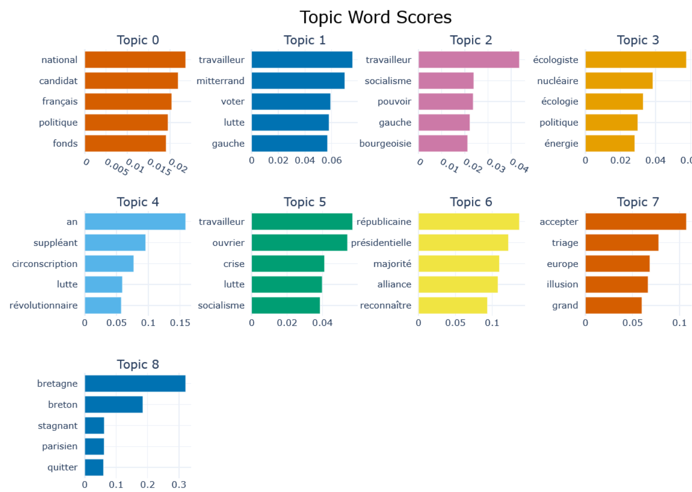

Figure 8: Top words for the BERTopic

Table 2: Synthesis of top words and topics for BERTopic via a LLM (in french)

Topic	**Mots dominants**	**Interprétation thématique**
Topic 0	national, candidat, français, politique, fonds	Politique nationale généraliste — discours de candidature, identité nationale. Topic dominant
Topic 1	travailleur, mitterrand, voter, lutte, gauche	Gauche mitterrandiste — campagne PS, monde du travail.
Topic 2	travailleur, socialisme, pouvoir, gauche, bourgeoisie	Gauche radicale / marxiste — vocabulaire de classe, bourgeoisie.
Topic 3	écologiste, nucléaire, écologie, politique, énergie	Écologie et nucléaire
Topic 4	an, suppléant, circonscription, lutte, révolutionnaire	Élections locales / extrême-gauche — vocabulaire électoral technique + révolutionnaire
Topic 5	travailleur, ouvrier, crise, lutte, socialisme	Lutte ouvrière / trotskisme — crise économique, monde ouvrier
Topic 6	républicaine, présidentielle, majorité, alliance, reconnaître	Institutions républicaines — majorité présidentielle, alliances politiques
Topic 7	accepter, triage, europe, illusion, grand	Europe et scepticisme
Topic 8	bretagne, breton, stagnant, parisien, quitter	Régionalisme breton - Centralisme vs régions

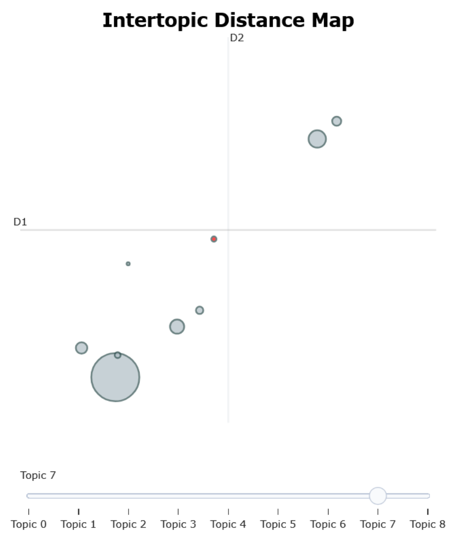

Figure 9: Projection of Bert topics in 2D

concrete thematic issues and seems better equipped to distinguish between different left-wing factions, although specific right-wing parties seem less explicitly defined.

Figure 9 shows the 2D UMAP projection of the topics center (centroids of the embeddings).The size of each circle is proportional to the number of documents in that topic. (An interactive version is available in the repository at `outputs/06_bertopic_intertopic.html`).

This distance map reveals the underlying semantic structure of the corpus. The space is largely dominated by Topic 0 in the bottom-left. It is a cluster that absorbs the generic national political discourse. Topic 6 is embedded within this giant cluster, indicating they act as semantic sub-categories of the mainstream discourse. More specific themes are logically isolated, such as Topic 8 (regionalism in Brittany) on the far left and Topic 7 (Europe) positioned distinctly in the center. Finally, while one could have intuitively expected all left-wing topics to form a single, tight semantic cluster, the projection reveals a more nuanced reality. Topics 1 (Mitterrandist Left) and 4 (Extreme-Left) form a completely distinct semantic pole in the top-right corner, far from the dominant cluster. One hypothesis which could explain this is that the vocabulary used by these left-wing parties was fundamentally distinct from the mainstream national political discourse captured by Topic 0. It may be interessant to investigate which parties are present in topic 1,2,4,5 since this 4 topics seem to be part of left-ring parties but are in the UMAP projection not close. This could say that each group has their own vocabulary/semantic approach.
Compared to the LDA projection which had a central and dense cluster with four indistinguishable topics, BERTopic demonstrates a superior ability to semantically discriminate between topics. Nevertheless it does not imply that these topics are more relevant that the LDA topics.

Dynamic topic modeling with BERTopic BERTopic allows us to track topic evolution over time using the local c-TF-IDF method. Figure 10 displays the number of documents assigned to each topic per election year.

The plot shows that Topic 0 remains the dominant discourse across all years with a rise between 1978 and 1981 before becoming more stable. Topics 1, 2, and 5 become progressively less dominant, with Topic 5 disappearing entirely after 1981. Topic 3 (Ecology) gains visible traction in 1981, which is historically relevant given the rise of political ecology in France during the 1970s. Certain topics do not appear as continuous lines on this plot because they are only present in a single election year. For instance, Topic 7 (Europe) appears only in 1988 (aligning with the growing European debate), while Topics 4, 6, and 8 (Bretagne) appear only in 1973. While this visualization can be susceptible to scaling issues due to the overwhelming volume of Topic 0, it effectively demonstrates the emergence of modern political themes such as ecology in the 80s or Europe at the end of the 80s and the decline

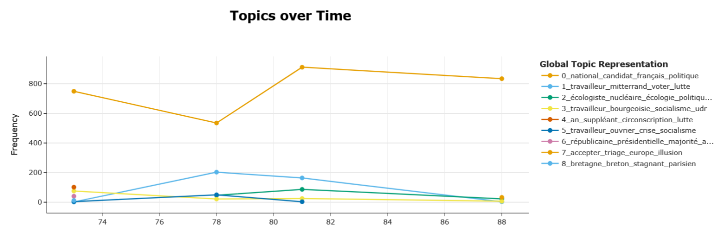

Figure 10: Evolution of BERTopic

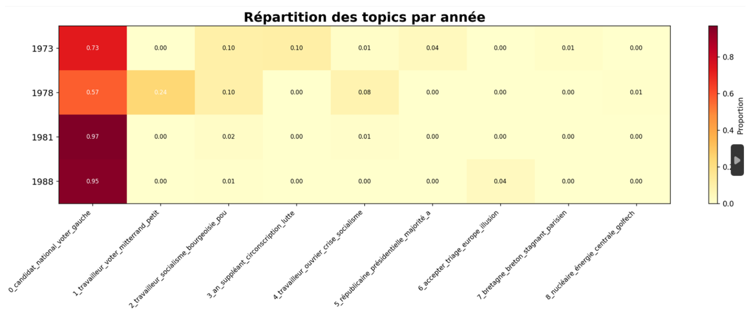

Figure 11: Heatmap distribution of BERT topics across years

of older, localized ones (regionalism in the 70s). Figure 11 shows a heatmap distribution of BERT topics across the year, which may be more easy to use to follow topics evolution.

Coherence metrics Figure 12 compares the automated coherence metrics between LDA and BERTopic, calculated using the top 10 words of each topic.

The C_{npmi} and U_{mass} metrics are slightly better for LDA than for BERTopic. This is expected, as these metrics evaluate strict word co-occurrence within documents which is in favor of Bag-of-Words algorithms like LDA. Indeed BERTopic relies on semantic embeddings, grouping conceptually related words even if they do not perfectly co-occur. This perfectly illustrates the "validation gap" highlighted by Hoyle et al. [2021], who demonstrated that traditional co-occurrence metrics are not necessarily suited for evaluating neural models. Furthermore, the absolute scores remain quite low across both models (ideally, U_{mass} should be closer to 0 and C_{npmi} strictly positive and higher). This reflects the difficulty of the corpus: the political vocabulary is highly homogeneous, making strict discrimination difficult. The C_v metric, generally considered the most aligned with human judgment, gives a slight edge to BERTopic.
Furthermore, it is important to note that the primary objective of this project was not to conduct

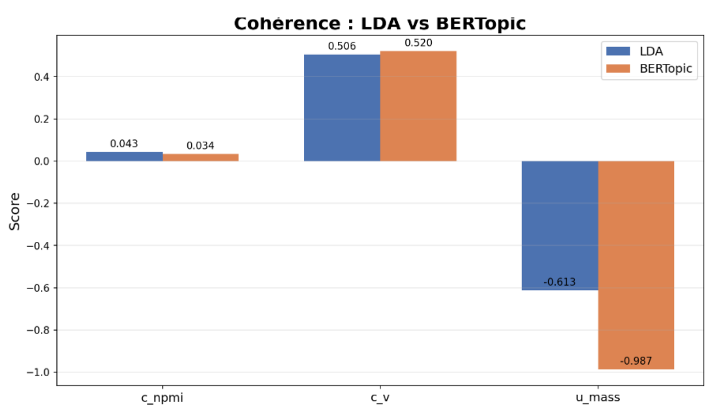

Figure 12: Coherence metrics

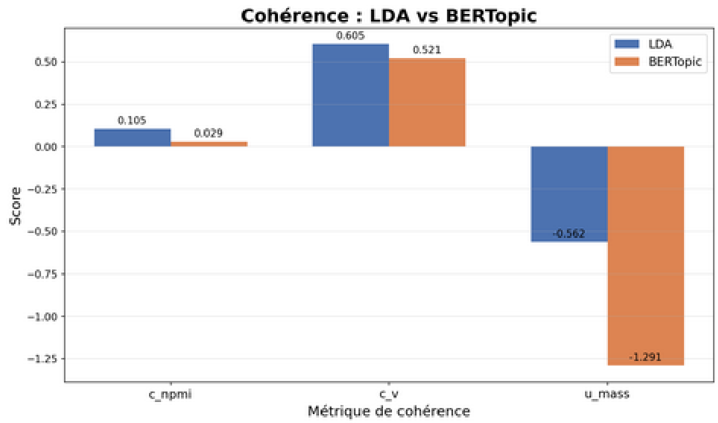

Figure 13: Coherence metrics with better hyper parameter

exhaustive hyperparameter tuning to achieve the absolute best topics, but rather to compare two distinct topic modeling methodologies and evaluate their dynamic topic modeling capabilities. Nevertheless, as illustrated in Figure 13, optimizing the parameters can lead to better coherence scores. For instance, adjusting the vocabulary filters to `n_features` = 2000 (instead of 1000), `min_df` = 3 (instead of 5), and `max_df` = 0.75 (instead of 0.9) yielded improved results.

5 Conclusion

While the quantitative coherence scores between the two models are relatively similar, a qualitative analysis demonstrates the superiority of BERTopic for this specific corpus. LDA serves as a solid baseline and successfully captures the main ideological structure (the left/right divide). However, it struggles with overlapping topics, forms dense and indistinct clusters, and even treats data artifacts as unique topics. In contrast, BERTopic provides added value. Thanks to its semantic embeddings, it uncovers concrete, distinct thematic issues that LDA entirely missed—such as ecology, European debates, and regionalism. Furthermore, it effectively manages noise by absorbing artifacts into outliers and offers much clearer semantic separation. Most importantly, it allows us to track the evolution of these themes over time, successfully fulfilling the primary objective of implementing a Dynamic Topic Modeling approach.

While this study successfully demonstrated the qualitative and structural advantages of BERTopic over LDA, the quantitative coherence scores remain relatively low for both models. This highlights a limitation of our current pipeline: the models were evaluated using random configurations rather than exhaustive hyperparameter optimization. Future work could involve a systematic grid search or Bayesian optimization to fine-tune parameters (such as the vocabulary size, UMAP dimensionality reduction parameters, or HDBSCAN cluster sizes). Furthermore, exploring different pre-trained transformer models fine-tuned specifically on historical or political French text could further enhance the semantic representation of the embeddings and yield even more distinct topics. It could be also interesting to have a look at each party individually in order to interpret the left-wing cluster better.

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